Sample: site and customer names removed · www.solaress.org

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DC EV charging stations, construction plan

Construction plan for four DC fast-charging stations · drawings in mm · BOM to follow

Site	Power	Chargers	Bays	Layout	New transformer
Amphoe Mueang Lamphun Standalone charging hub · Highway 114	240 kW	2 × 120 kW	4	Row + canopy	315 kVA
Market car park, Mukdahan Market car park · next to fuel station	360 kW	3 × 120 kW	6	Row + canopy	500 kVA
Community mall, Ubon Ratchathani Community mall car park	360 kW	4 × 90 kW	8	Row + canopy	500 kVA
Roadside lot, Nakhon Ratchasima Near a university · empty lot	540 kW	3 × 180 kW	6	Centre island + T-canopy	630 kVA

Common layout rules

- EV bay 2,600 × 5,500 (wider than a normal bay for cable reach)
- One dual-gun charger serves 2 bays; charger cable 4–5 m
- Chargers on a raised island +150 with plinth +200, 2 bollards each
- Drive aisle ≥ 6,000 two-way, ≥ 2,500 one-way per side
- Canopy underside ≥ 3,200 above floor

Electrical concept

- New PEA transformer 22 kV / 400-230 V per site, in a fenced yard or kiosk
- Outdoor MDB IP54 with main breaker, SPD Type 1+2, one MCCB per charger, energy meter
- Copper XLPE cables in HDPE ducts, trench 1,000 deep under driveways
- TN-S earthing, ring of rods, ≤ 5 Ω
- Each charger has its own RCD and emergency stop

Basis and limits

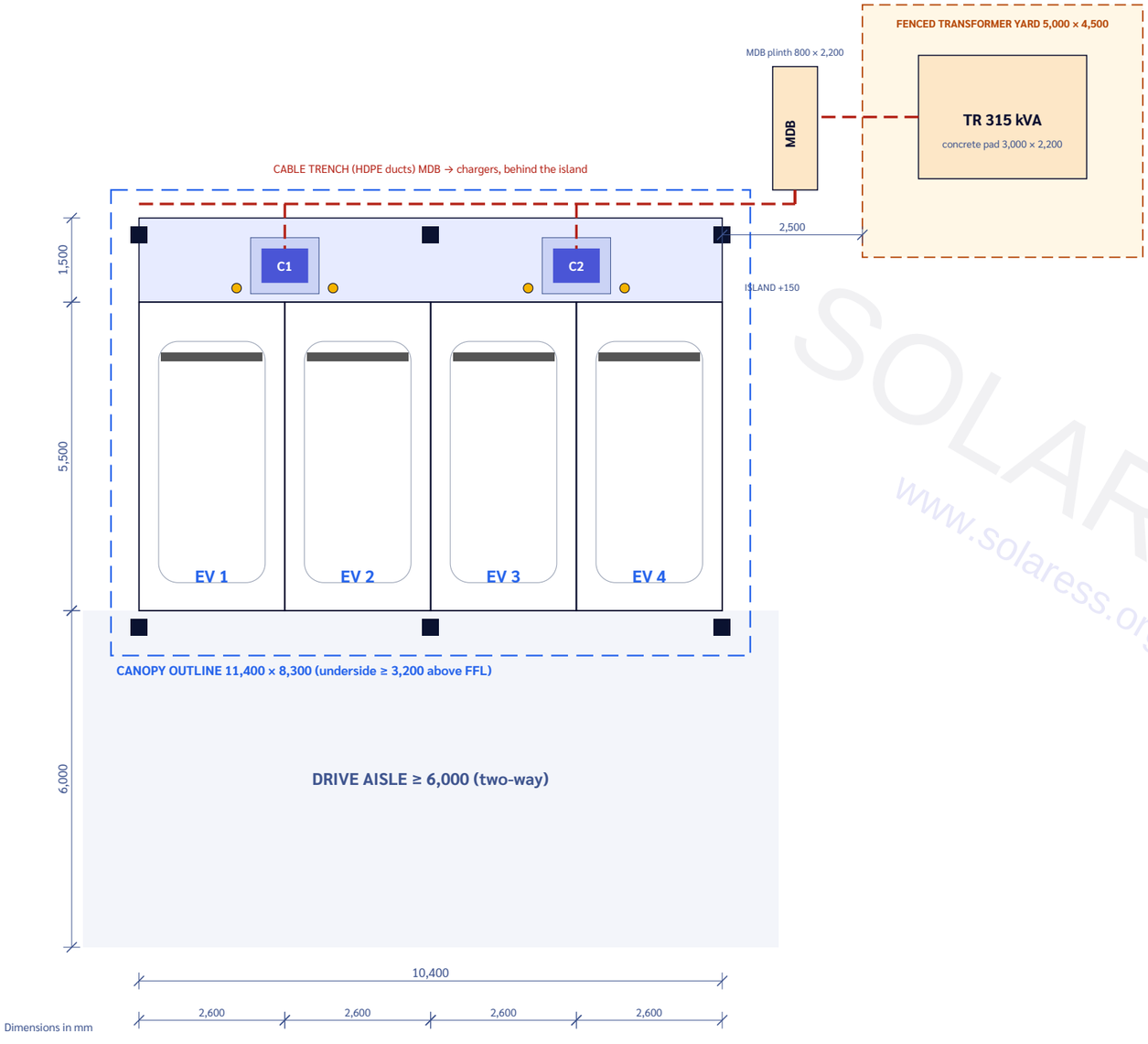
- Taken from the four project decks: renders, bay counts, charger datasheets and the Nakhon Ratchasima landlord sketch
- No site survey yet: all positions are to be confirmed on site
- Structure and footings are indicative only, to be designed by licensed engineers
- Transformer size assumes all chargers at full power at once

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1. Site plan, Amphoe Mueang Lamphun

Standalone charging hub · Highway 114 · 240 kW · 4 bays · plan scale 1:100 on A3



Design basis

Station power	240 kW DC
Chargers	2 × 120 kW, dual CCS2 (2 bays each)
Bays	4 EV bays, 2,600 × 5,500 mm
Charger footprint (W × D × H)	820 × 600 × 1,950 mm
Input per charger	3-phase 400 V, ≈ 185 A
Breaker per charger (MDB)	250 A 3P
Cable per charger (Cu XLPE, in duct)	4 × 1C 120 mm² + 1C 70 mm² PE
Transformer (new, PEA)	315 kVA 22 kV / 400-230 V
MDB main breaker	MCCB 630 A 3P
Transformer → MDB	4 × (2 × 1C 240 mm²) + PE 120 mm²

Site notes

- Open lot on Highway 114: set out from the property boundary and road reserve; keep the row ≥ 3,000 inside the boundary.
- Raise FFL ≥ 150 above the road crown; slope 1 % away from chargers to a drain.
- Transformer yard at the end of the row nearest the PEA 22 kV line (confirm pole position).

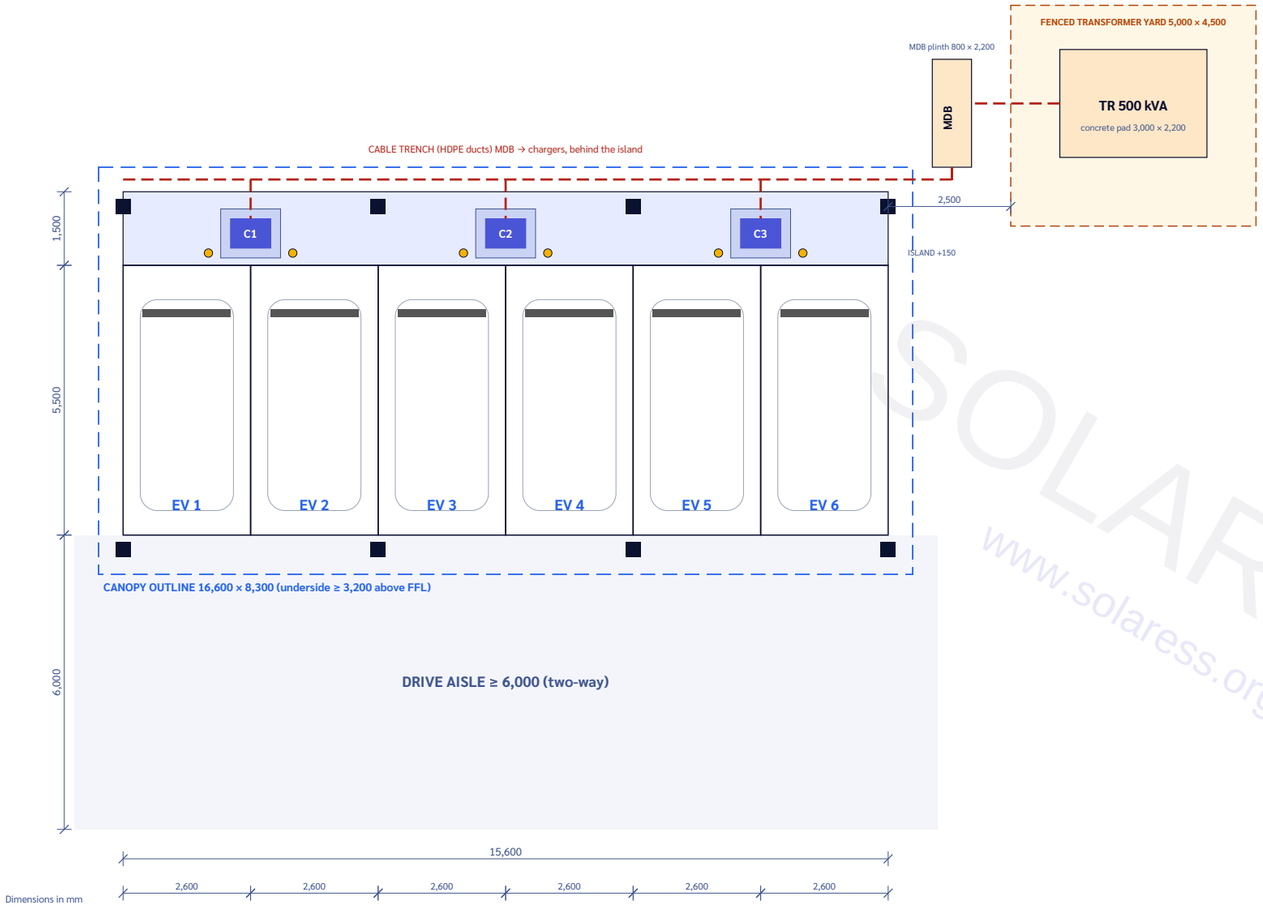
Charger · Plinth · Canopy column · Bollard · Cable trench

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2. Site plan, Market car park, Mukdahan

Market car park · next to fuel station · 360 kW · 6 bays · plan scale 1:125 on A3



Design basis

Station power	360 kW DC
Chargers	3 × 120 kW, dual CCS2 (2 bays each)
Bays	6 EV bays, 2,600 × 5,500 mm
Charger footprint (W × D × H)	820 × 600 × 1,950 mm
Input per charger	3-phase 400 V, ≈ 185 A
Breaker per charger (MDB)	250 A 3P
Cable per charger (Cu XLPE, in duct)	4 × 1C 120 mm² + 1C 70 mm² PE
Transformer (new, PEA)	500 kVA 22 kV / 400-230 V
MDB main breaker	ACB 800 A 3P
Transformer → MDB	4 × (3 × 1C 240 mm²) + PE 185 mm²

Site notes

- Market already has 4 EV bays: keep the new row separate and keep them running during works.
- Fuel station next door: keep chargers, MDB and transformer outside its hazardous zone (check distance with the fuel station and PEA).
- Night work or barriers needed: the market is busy all day.

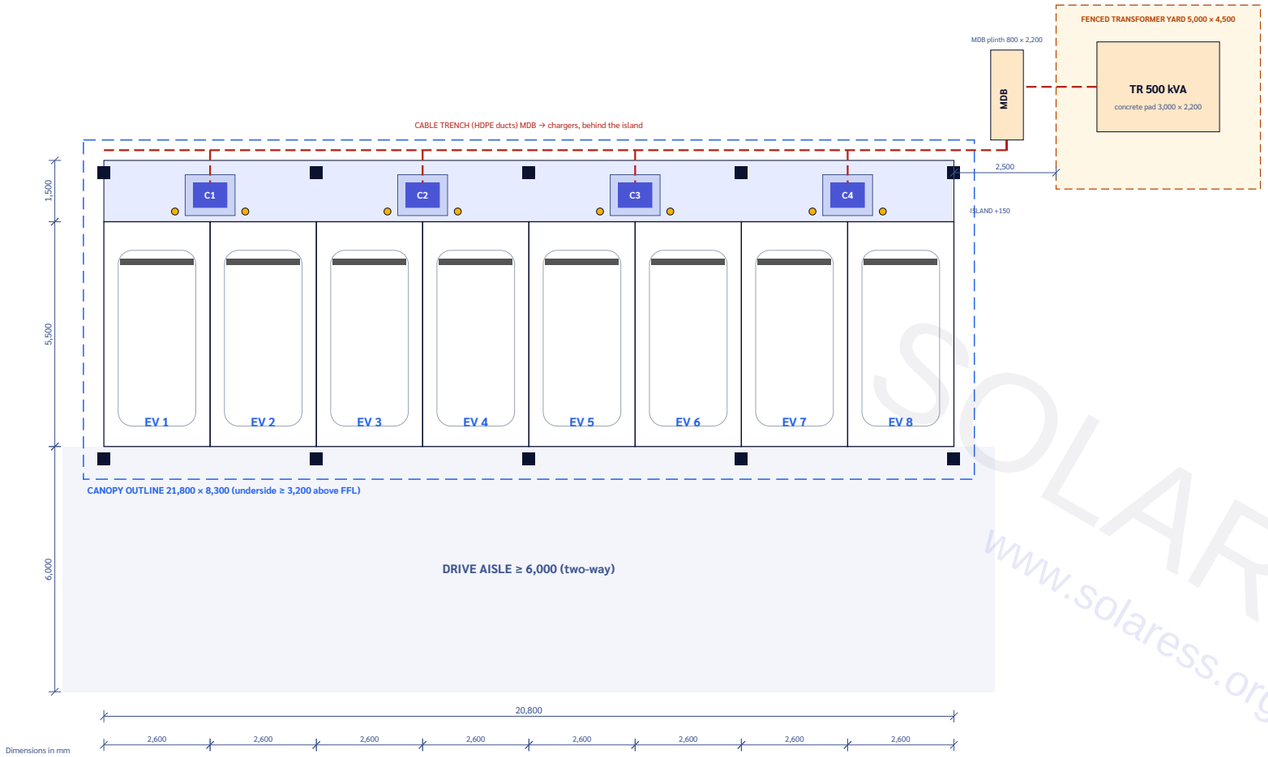
Charger · Plinth · Canopy column · Bollard · Cable trench

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3. Site plan, Community mall, Ubon Ratchathani

Community mall car park · 360 kW · 8 bays · plan scale 1:150 on A3



Design basis

Station power	360 kW DC
Chargers	4 × 90 kW, dual CCS2 (2 bays each)
Bays	8 EV bays, 2,600 × 5,500 mm
Charger footprint (W × D × H)	820 × 600 × 1,950 mm
Input per charger	3-phase 400 V, ≈ 140 A
Breaker per charger (MDB)	160 A 3P
Cable per charger (Cu XLPE, in duct)	4 × 1C 70 mm² + 1C 35 mm² PE
Transformer (new, PEA)	500 kVA 22 kV / 400-230 V
MDB main breaker	ACB 800 A 3P
Transformer → MDB	4 × (3 × 1C 240 mm²) + PE 185 mm²

Site notes

- Mall car park (opens 08:00–02:00): fence the work area, work in phases, keep fire lanes open.
- Check existing underground services before cutting the asphalt (drains, lighting cables).
- The deck render shows 4 chargers for 8 bays: this plan uses 4 × 90 kW = 360 kW. Confirm the model.

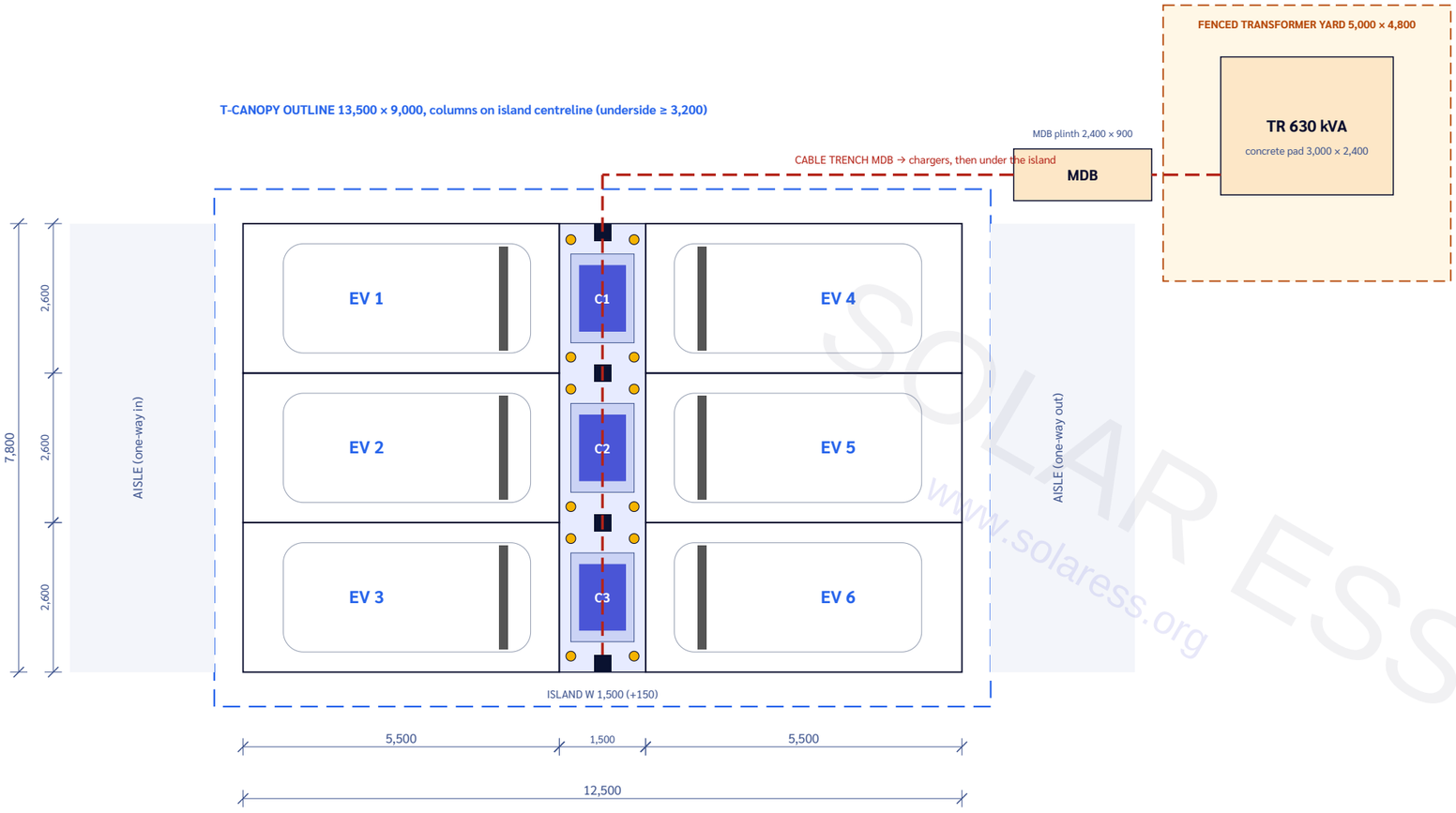
Charger · Plinth · Canopy column · Bollard · Cable trench

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4. Site plan, Roadside lot, Nakhon Ratchasima

Near a university · empty lot · 540 kW · 6 bays · plan scale 1:100 on A3



Landlord sketch: bays 2,500 × 5,000, island 800. This plan widens the island to 1,500 for 180 kW chargers (W 750–1,145, D 720–800) with bollards.

Dimensions in mm

Design basis

Station power	540 kW DC
Chargers	3 × 180 kW, dual CCS2 (2 bays each)
Bays	6 EV bays, 2,600 × 5,500 mm
Charger footprint (W × D × H)	1,145 × 800 × 2,196 mm
Input per charger	3-phase 400 V, ≈ 277 A
Breaker per charger (MDB)	400 A 3P
Cable per charger (Cu XLPE, in duct)	4 × 1C 185 mm² + 1C 95 mm² PE
Transformer (new, PEA)	630 kVA 22 kV / 400-230 V
MDB main breaker	ACB 1,000 A 3P
Transformer → MDB	4 × (4 × 1C 240 mm²) + PE 240 mm²

Site notes

- Empty lot: clear, cut and fill, compact in 200 mm layers to 95 % modified Proctor, then 150 mm crushed rock + slab.
- T-canopy with columns only in the island (as in the landlord sketch and render); its 6.25 m cantilever needs an engineer's design.
- Entrance and exit as the landlord sketch: one-way loop around the island.

Charger · Plinth · Canopy column · Bollard · Cable trench

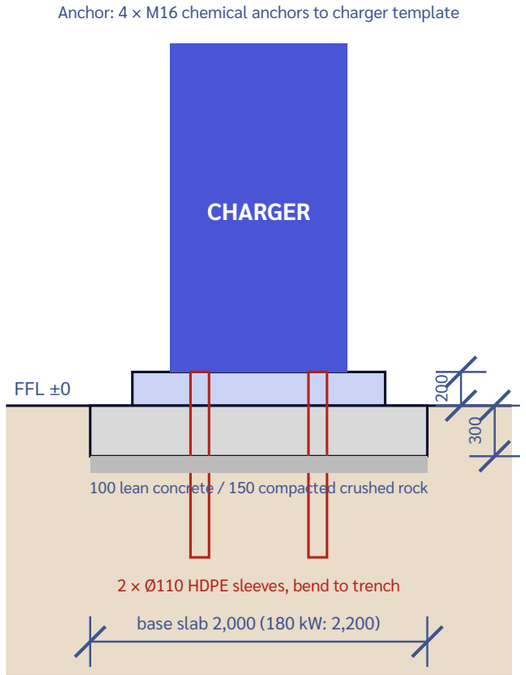
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Typical construction details

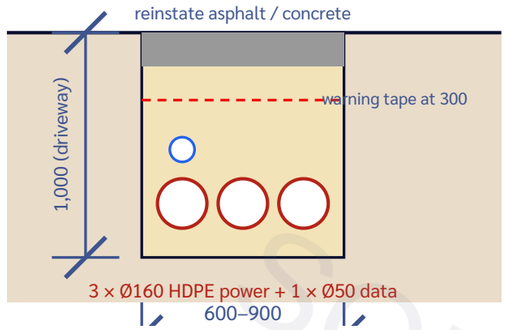
Typical details for all four sites · dimensions in mm · not to scale between details

A. Charger plinth (section)



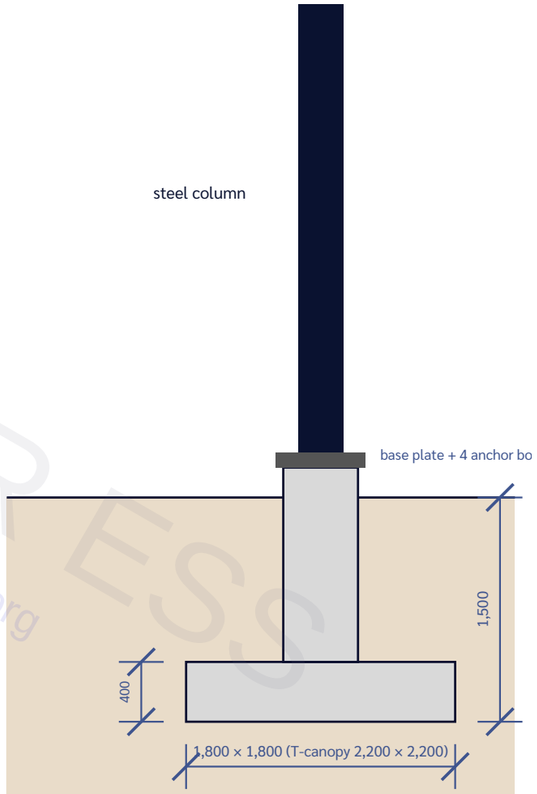
Reinforced concrete 240 ksc, DB12 @200 both ways. Plinth 200 high, ≥ 200 wider than the charger each side. Level ± 2 mm.

B. Cable trench (section)



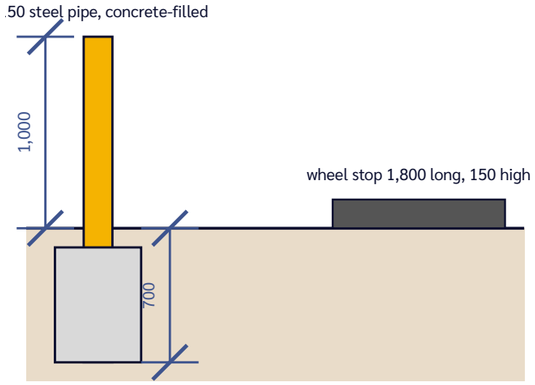
Sand bed 100 under ducts, sand surround to 150 above, backfill compacted in 200 layers. Pull pits 600 × 600 at bends and every ≤ 25 m.

C. Canopy column footing (indicative)



Pad footing on firm soil (≥ 10 t/m², check with a soil test). Column, footing and anchor sizes to be designed and signed by a licensed civil engineer.

D. Bollard and wheel stop



Two bollards per charger, 450 from the charger edge. Wheel stop 900 from the island face so bumpers cannot touch the charger.

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Construction sequence and programme

One site, about 12 weeks from survey to handover. Sites can run in parallel with one crew per site.

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Survey and design Topographic and utility survey, soil test (1 borehole or DCP per canopy), set-out, final drawings, engineer signature												
Permits and PEA Building permit for the canopy (local authority), PEA application for new transformer and meter; PEA lead time can be longer												
Site set-up and safety Hoarding, signs, traffic plan, site office, CAR insurance notified before start												
Earthworks and footings Cut asphalt, excavate, footings, anchor bolts, curing 7 days before loading												
Trenches, ducts, earthing HDPE ducts, pull pits, earth rods and ring, earth test ≤ 5 Ω												
Plinths, island, slab Island kerb, charger plinths, transformer pad, MDB plinth, paving												
Canopy steel and roof Erect columns and beams, roof sheet, gutters, downpipes, LED lights												
Transformer and MDB PEA sets transformer (or contractor under PEA inspection), MDB set and terminated												
Cables and chargers Pull and terminate cables, set chargers on plinths, data network / 4G												
Testing and energising IR test, continuity, polarity, RCD, earth, torque check, PEA inspection and meter, energise												
Commissioning and handover Charge test each gun, OCPP to backend, line marking, signs, as-built drawings, handover												

Hold points (sign before going on)

- Set-out checked against the drawing
- Footing excavation and rebar before pouring
- Ducts and earthing before backfill
- Anchor bolt position before setting chargers
- Insulation and earth tests before energising

Standards and approvals

- Electrical work to the EIT wiring standard and PEA rules for EV chargers
- Chargers IEC 61851-1 / -23, CCS2
- Canopy designed and signed by a licensed civil engineer
- Electrical design signed by a licensed electrical engineer
- Fire extinguisher and emergency stop sign at each island

Safety on site

- Barrier the work area from public traffic at all times
- Work at height plan for canopy erection
- Lock-out tag-out before any connection
- Daily toolbox talk; PPE for all workers
- Each site notified to the CAR insurer before start

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Items to confirm

Answer these before the final drawings and the BOM

All sites

- Exact charger model and rating per site (decks list three brands)
- Who supplies the transformer: PEA or the developer
- Location of the nearest PEA 22 kV line and pole
- Lease area boundaries and where the transformer may stand
- Canopy style: the render shows a 2-row column canopy; confirm brand colours and height

Lamphun 240 kW

- Site levels against Highway 114 and drainage outlet
- Road reserve and setback from the highway

Mukdahan 360 kW

- Distance from the fuel station dispensers and tanks
- Whether the existing 4 bays share the same supply

Ubon 360 kW

- 4 × 90 kW or another split; the render shows 4 chargers for 8 bays
- Mall works hours and fire-lane rules

Nakhon Ratchasima 540 kW

- Accept widening the island from 800 to 1,500
- Soil for fill and T-canopy footings (soil test)
- Final entrance and exit as the landlord sketch

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